

Inchcolm Island & Forth Bridge (Kids)

UK Family Tour

The Big Red Bridge Made of 6.5 Million Rivets

Before your boat even leaves, look up at the giant red bridge marching across the water on huge steel legs. That's the Forth Bridge, opened in 1890, and it was the biggest thing humans had ever built out of steel — fifty-three thousand tonnes of it, held together by about six and a half million rivets, each one heated glowing-hot and hammered in by hand. It took seven years and about four and a half thousand workers. The last rivet, a gold-plated one, was tapped in by the future King Edward the Seventh. Why does it look so massively overbuilt? Because eleven years earlier, a big Scottish railway bridge over the River Tay collapsed in a storm while a train was crossing, and everyone aboard died. Britain was so shocked that the Forth Bridge's engineers made it not just strong, but visibly, obviously, unmistakably strong — so every passenger could see with their own eyes that it would never fall. Trains still rumble across it today, about two hundred every day, one hundred thirty-six years later. In Britain, when a job seems to never end, people say it's "like painting the Forth Bridge" — because painting it took so long that when crews finished, legend says, they had to start again at the other end. Your boat sails right underneath it. Wave to the trains.

The Three Men and the Chairs

Here is how the engineers proved the bridge would work, and you can try it yourself later. Before the bridge was even finished, a clever engineer named Benjamin Baker wanted to show everyone that its funny diamond shape was actually incredibly strong. So he did something brilliant. He sat two grown men on two chairs, a little way apart, and told each of them to stretch out their arms and hold a broomstick, with a big heavy pile of bricks stacked beside them. Then, in the empty space between the two men, held up by nothing but their outstretched arms, a third man sat down in mid air. And he did not fall. That is the whole secret of the Forth Bridge, shown with three people. The two men on the chairs are the bridge's towers. Their stretched-out arms, pulling hard, are the steel bars along the top. Their bodies and the broomsticks, pushing and squeezing, are the

giant steel tubes along the bottom. And the piles of bricks are the heavy ends that anchor the whole thing down so it can never tip over. The man floating in the middle? That's the train, crossing safely. Now for the best part. The man they chose to sit in the middle was a young engineer named Kaichi Watanabe, who had travelled all the way from Japan to help build the bridge. They picked him on purpose, as a thank you to the countries of the East, where builders had used this clever reaching-arm idea for hundreds and hundreds of years. So next time you are with two grown-ups and a few cushions, give it a go. Two people on chairs with their arms stretched out, and a smaller person perched carefully between them. Can you feel how the arms have to pull while the bodies push? The clever thing is that the two men are not really lifting the weight at all. They are pulling and pushing against each other, and against their piles of bricks, and that quiet battle of pulling and pushing is exactly what holds the middle man up in the air. A real bridge does the same trick, only with steel instead of arms. If nobody wobbles over, congratulations. You have just turned yourselves into the Forth Bridge.

How Tall, How Long, How Heavy

Let's play a game of how big. Look at the two enormous arches that make up the middle of the bridge. Each one reaches more than five hundred metres across the water, which is longer than five football pitches laid end to end, all in a single jump, with no legs holding up the middle. When the bridge opened in 1890, no bridge anywhere in the whole wide world could leap further in one go. It stayed the world champion for twenty seven whole years. Now look at how high up the train track is. It runs about forty six metres above the sea, as tall as a building with fifteen floors. The builders made it that high on purpose, so that the tallest sailing ships and warships could glide underneath without their masts bumping the bottom. And here is a sneaky secret. The legs in the very middle of the bridge are not standing on the seabed at all. They are standing on a whole little island, called Inchgarvie, sitting right in the middle of the water, which the builders used like a giant stepping stone. Then

there is the paint. For more than a hundred years, painting this gigantic bridge took so long that the very moment the painters reached the far end, they had to go straight back to the start and begin all over again, forever and ever. That is why people used to say a job that never ends is like painting the Forth Bridge. But not anymore. A while ago the bridge got a coat of super-tough, space-age paint, the same kind that protects oil platforms far out at sea, and it should last for twenty five years or more. So the painters, at long last, after more than a century of climbing about in the wind and the rain, finally got to go home. One more number to boggle your brain. The whole bridge is held together by about six and a half million rivets, which are like giant metal buttons, and if you tried to count them out loud, one every second, without ever stopping to sleep or eat, it would take you more than two months. As you sail underneath, give the old bridge, and its tired painters, a wave too.

Three Bridges in a Row

As your boat slips through, do a quick count. How many bridges cross the water here? The answer is three, and they are like three brothers of very different ages, standing in a line. The nearest one, the big red one made of steel diamonds, is the oldest brother. It's the railway bridge, built in 1890, and only trains are allowed to cross it. Next along is a bridge that looks completely different, hanging from two tall towers on long swooping cables, a bit like a giant skipping rope stretched across the water. That is the road bridge, built in 1964, for cars and buses and lorries. And then there is the newest brother of all, gleaming and white, with three pointed towers and cables fanning out like the strings of a harp. That one opened in 2017. So the oldest bridge here and the newest bridge here were built a hundred and twenty seven years apart, yet they stand right next to each other, sharing the same view. Why build three bridges in one place? Well, first came the trains, so people built a bridge just for trains. Then along came motorcars, and cars cannot drive on a railway, so they built a second bridge just for the road. But after fifty years of heavy lorries thundering across, the road bridge grew tired, and its thick cables started to wear out deep inside, where nobody could see. So rather than risk it, they built the beautiful white bridge to take over the job. Three bridges, three different jobs, three different times, all crossing the exact same stretch of water. See if you can point to all three, and remember the trick. Red is the oldest, and white is the newest, with the cars in between. Here is a game for the boat. Imagine you could

travel in time just by hopping from one bridge to the next. Step onto the red one and you are back in the days of steam trains and top hats. Step onto the middle one and it is the nineteen sixties. Step onto the white one and you are almost in today. Three bridges, and each one is like a doorway into a different age.

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The Island of the Monks

Inchcolm is called the "Iona of the East" — a tiny holy island where monks lived for hundreds of years. The story begins in 1123, when King Alexander the First of Scotland was caught in a violent storm crossing the Forth and shipwrecked here. A hermit — a monk living alone on the island in a tiny cell, surviving on shellfish and the milk of one cow — sheltered the king for three days. The grateful king promised to build a monastery on the island, and Inchcolm Abbey was born. Because it's on an island, the abbey escaped the destruction that ruined most Scottish monasteries, so it's the best-preserved one in Scotland — you can climb the tower's narrow spiral stairs, explore the echoing chapter house with its stone dome, and creep along the actual dark cloister passages where monks walked, prayed and worked eight hundred years ago. Try the echo in the chapter house — one clap. Monk life was tough: they woke in the middle of the night for prayers, every night, forever. But island life had other problems too: pirates! English raiders attacked the island monks several times, stealing everything, including once — the story goes — a beloved statue. Legend says the raiders' ship promptly hit trouble, and the frightened sailors brought it back. Don't mess with the island.

The Island in a Shakespeare Play

Here is something almost nobody knows about this tiny island. It has a starring role in one of the most famous plays ever written. About four hundred years ago, a writer named William Shakespeare wrote a spooky, thrilling play called *Macbeth*, all about Scottish kings and witches and battles in the dark. And right near the beginning, he mentions a real island by name. This one. In the play, an army of invaders from Norway has just lost a huge battle against the Scots. Their soldiers lie dead on the ground, and the Norwegian king wants to bury them properly. But the Scots say no, not until you pay us first. And where do they make the beaten Norwegians hand over their treasure, ten thousand coins of it, before they are allowed to bury their dead? At Saint Colme's Inch, which is the old name for this very island, the island of Saint Columba. Now, that particular story is

probably just a legend, not something that truly happened. But real invaders certainly did come here long ago. Fierce Viking raiders sailed these grey waters in their longships and attacked little holy islands exactly like this one. And here is the funny twist that ties it all together. Hundreds of years ago, a monk who actually lived right here on Inchcolm wrote an enormous history book about Scotland, full of kings and battles. Bits of that book were copied out and passed along, year after year, until at last they reached Shakespeare himself, and helped give him the ideas for his play. So a monk on this quiet rock helped spark a story, and the story ended up on stages all over the world, with the island's own name spoken out loud inside it. Imagine that. An actor on a stage in faraway London, saying the name of the little island you are sailing toward right now. And here is a bonus puzzle for the play. Shakespeare says the Norwegians paid in dollars, but dollars had not even been invented yet in those ancient days, so he got that bit wrong by hundreds of years. Writers make mistakes too. But he got one thing exactly right. This island really was a famous, special place where important people longed to be buried, close to a holy saint. So when you land, you are stepping onto a spot that a great playwright thought worth putting into a play.

Secrets to Find in the Ruins

Let's go on a treasure hunt through the abbey, because this is no ordinary ruin. Most old Scottish monasteries are just broken, tumbledown walls now, but this one still has real rooms with real roofs that you can walk right inside, and it hides some brilliant secrets. Secret number one, find the eight-sided room called the chapter house. It still has its stone roof, and it makes an amazing echo. Stand in the middle, say one word or clap once, and listen to it bounce around the stone. This is where the monks sat every single morning to be given their jobs for the day. Secret number two, walk along the covered stone corridors that run around the square garden. They are called the cloister, and this is the most complete cloister left anywhere in Scotland, the very same sheltered walkways the monks paced up and down eight hundred years ago. Secret number three, and this is the best of all, look for a real painting on the wall, tucked into a stone nook, that is more than seven hundred years old. It shows a line of people in hoods, walking slowly in a funeral. Once upon a time, almost every wall in here was covered in bright colours and pictures, and this little faded patch is one of the only bits that survived. Secret number four, find the tiny stone hut where a hermit once lived all

alone with just one cow, because that lonely little hut is how the whole abbey began. And finally, if your legs are feeling brave, climb the twisty tower stairs all the way to the top for the best view of the entire day. Here is the big question to solve. Why did this abbey survive when so many others were smashed to pieces? The answer is easy. It sits on an island. When people came along wanting to knock the monasteries down and carry off the stones, they took one look at all that cold water in the way, and decided it simply was not worth the boat trip.

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Seals, Seabirds and Secret War Tunnels

Keep watch on the crossing: these waters are full of wildlife. Grey seals love to lounge on the rocks and buoys near the island like fat bananas — the boat crew usually knows where they're hanging out today, and you might see their dog-like heads pop out of the water to inspect your boat. In early summer, puffins zip past like flying potatoes with orange feet. One warning about July: it's seagull chick season. Parts of the island are guarded by nesting gulls who dive-bomb anyone who walks too close to their fluffy chicks. Follow the marked paths, and if a gull swoops you, raise your hand or a hat above your head — they aim for the highest point. Consider it the island's air force. Now here's the surprise: this holy island of peaceful monks is also a fortress. Because Inchcolm sits right in the sea lane to Edinburgh and the naval bases, it was armed to the teeth in both World Wars — gun batteries, searchlights, barracks, and a network of tunnels cut through the island's spine that you can walk through today. A hundred soldiers lived here where monks once prayed. Explore the dark tunnel — it's genuinely spooky — then have your picnic on the beach where, eight hundred years apart, both monks and soldiers watched the same water for trouble.

The Soldiers' Island

You already know that this peaceful island of monks once turned into a fortress. But let's find out what the soldiers actually did here, because it is a proper adventure story. The soldiers first arrived in 1915, during the First World War, and they had one big, important job. They had to guard the doorway to the river, so that enemy ships and sneaky submarines could not slip upstream and attack the navy. To do it, they packed this little island with eight guns that could fire really, really fast, some of them right up near the top of the hill, pointing in every direction so that nothing could possibly get past. At night, they switched on enor-

mous searchlights, far brighter than any car headlight, and swept the beams across the black water, hunting for the shadow of an enemy boat trying to creep by in the dark. They even stretched a giant net, called a boom, all the way across the water between the islands, like an underwater fence to trap submarines. And here is the really clever part. To carry heavy boxes of ammunition to the guns without being seen, they dug a tunnel straight through the middle of the hill, so the soldiers could hurry through in secret from one side to the other. That is the dark, spooky tunnel you can walk through today, so remember, real soldiers once rushed along it carrying shells. About a hundred soldiers lived out here on this windy little island, sleeping in huts, keeping watch over the water day and night, in the very same place where, hundreds of years before them, gentle monks had knelt down to pray. Same island, same grey view, but a completely different job. Think how strange it must have been. In the daytime the soldiers could look up and see the ruined abbey, all quiet and holy, and at night they stood beside their guns in the freezing wind, listening for the rumble of an enemy engine out on the water. As you explore, see if you can spot where the big guns once stood, and where the searchlights once blazed out across the dark.

Where the Animals Live

The Firth of Forth is like a huge wildlife motorway, and the islands around you are the busy stopping places where the animals rest and raise their babies. Take the grey seals you might spot lolling about on the rocks. In summer they are mostly just relaxing and fishing, but in the autumn they gather in enormous crowds on quiet islands and beaches

to have their pups. And a baby seal is born wearing a coat of fluffy white fur, so soft and fluffy that it cannot even swim at first. Its mother feeds it thick, creamy milk until it grows fat and round, and only then does it lose the white fluff, grow a sleek waterproof coat, and finally slip into the sea. Then there are the puffins, those little birds with the bright rainbow beaks. Puffins do not build nests up in trees. Instead they dig burrows down in the ground, like rabbit holes, on grassy islands nearby, and they only visit in the summer to raise a single chick, which has the wonderful name of a puffling. Once summer ends, they fly far, far out to sea and stay there all winter long. Further out toward the open water is a very special place called the Bass Rock, a whole island that looks as though it is covered in snow. But it is not snow at all. It is tens of thousands of big white birds called gannets, so many of them squashed together that they turn the whole rock white. It is the biggest gathering of gannets on the entire planet. And if you keep your eyes on the waves, you might even catch a dolphin or a porpoise rolling past. So watch the water carefully. Every splash, every bobbing head, every whirring pair of wings could be one of the island's neighbours, heading home. And here is why all these creatures crowd into the same corner of the sea. The Firth of Forth is packed with fish, which is a giant free dinner for seals and puffins and gannets alike, and the little rocky islands make safe places to rest and hide from danger. So this really is a wildlife city, with the fish as its food shops and the islands as its houses. You just happen to be visiting on ferry day.